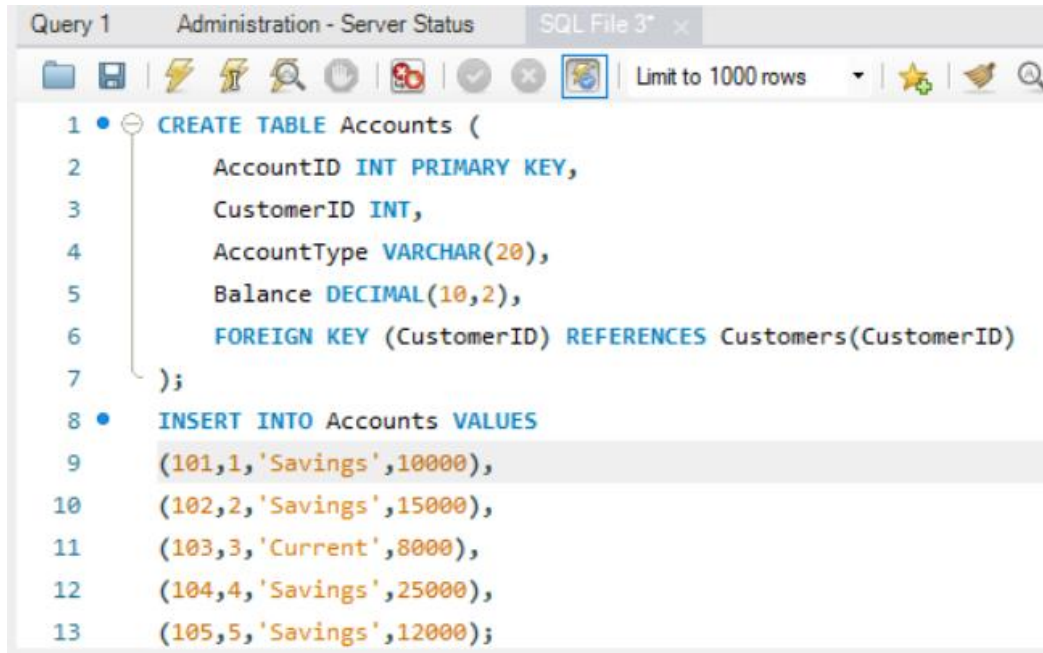


Week 2

Exercise 2

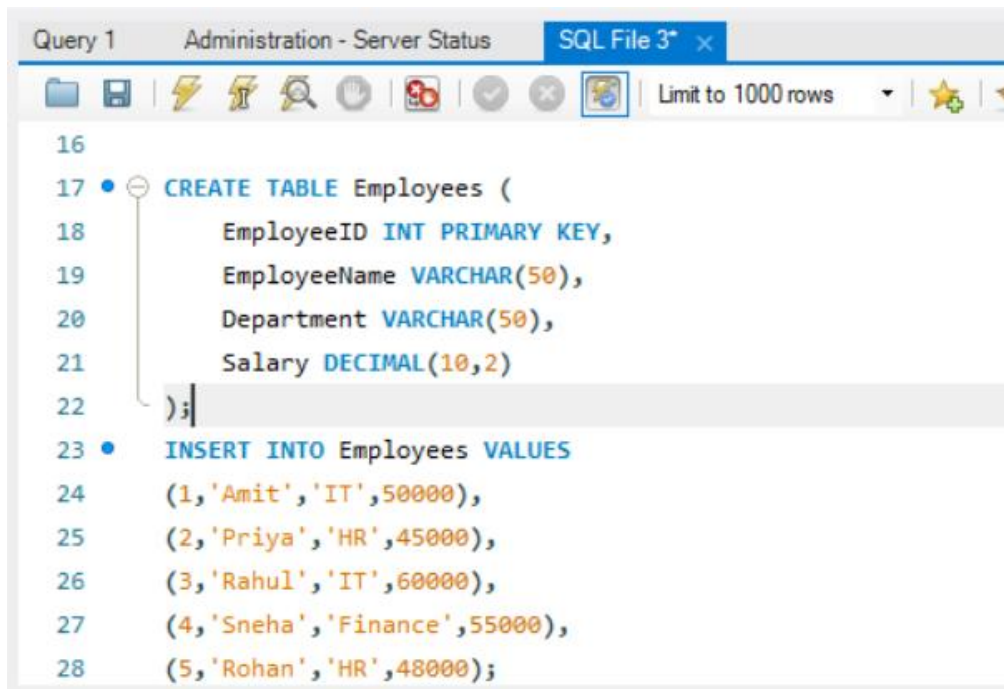
Created Accounts Table



The screenshot shows a SQL query editor with two tabs: 'Query 1' and 'Administration - Server Status'. The 'Query 1' tab is active, displaying SQL code for creating and inserting data into the 'Accounts' table. The code is as follows:

```
1 CREATE TABLE Accounts (  
2     AccountID INT PRIMARY KEY,  
3     CustomerID INT,  
4     AccountType VARCHAR(20),  
5     Balance DECIMAL(10,2),  
6     FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)  
7 );  
8 INSERT INTO Accounts VALUES  
9 (101,1,'Savings',10000),  
10 (102,2,'Savings',15000),  
11 (103,3,'Current',8000),  
12 (104,4,'Savings',25000),  
13 (105,5,'Savings',12000);
```

created Employees table



The screenshot shows a SQL query editor with two tabs: 'Query 1' and 'Administration - Server Status'. The 'Query 1' tab is active, displaying SQL code for creating and inserting data into the 'Employees' table. The code is as follows:

```
16  
17 CREATE TABLE Employees (  
18     EmployeeID INT PRIMARY KEY,  
19     EmployeeName VARCHAR(50),  
20     Department VARCHAR(50),  
21     Salary DECIMAL(10,2)  
22 );  
23 INSERT INTO Employees VALUES  
24 (1,'Amit','IT',50000),  
25 (2,'Priya','HR',45000),  
26 (3,'Rahul','IT',60000),  
27 (4,'Sneha','Finance',55000),  
28 (5,'Rohan','HR',48000);
```

Scenario 1 Process Monthly Interest

```
--  
33 DELIMITER //  
34  
35 • CREATE PROCEDURE ProcessMonthlyInterest()  
36 BEGIN  
37     UPDATE Accounts  
38     SET Balance = Balance + (Balance * 0.01)  
39     WHERE AccountType = 'Savings';  
40 END //  
41  
42 DELIMITER ;  
43
```

Calling the procedure and displaying

```
44  
46 • CALL ProcessMonthlyInterest();  
47  
48 • SELECT * FROM Accounts;
```

Result Grid	Filter Rows:	Edit:	Export/Import:
AccountID	CustomerID	AccountType	Balance
101	1	Savings	10100.00
102	2	Savings	15150.00
103	3	Current	8000.00
104	4	Savings	25250.00
105	5	Savings	12120.00
NULL	NULL	NULL	NULL

Scenario 2 Update Employee Bonus

```
53 • CREATE PROCEDURE UpdateEmployeeBonus(  
54     IN dept VARCHAR(50),  
55     IN bonusPercent DECIMAL(5,2)  
56 )  
57 BEGIN  
58     UPDATE Employees  
59     SET Salary = Salary + (Salary * bonusPercent / 100)  
60     WHERE Department = dept;  
61 END //  
62  
63 DELIMITER ;  
64
```

Calling and displaying

```
69  
70 • SELECT * FROM Employees;
```

Result Grid				
Filter Rows:				
Edit:    Export/Import: 				
	EmployeeID	EmployeeName	Department	Salary
▶	1	Amit	IT	55000.00
	2	Priya	HR	45000.00
	3	Rahul	IT	66000.00
	4	Sneha	Finance	55000.00
	5	Rohan	HR	48000.00
*	NULL	NULL	NULL	NULL

Scenario 3 Transfer Funds

DELIMITER //

```
CREATE PROCEDURE TransferFunds(  
    IN FromAccount INT,  
    IN ToAccount INT,  
    IN Amount DECIMAL(10,2)  
)  
BEGIN  
    DECLARE CurrentBalance DECIMAL(10,2);  
  
    SELECT Balance  
    INTO CurrentBalance  
    FROM Accounts
```

Query 1

Administration - Server Status

SQL File 3* x



```
85     FROM Accounts  
86     WHERE AccountID = FromAccount;  
87  
88     IF CurrentBalance >= Amount THEN  
89  
90         UPDATE Accounts  
91         SET Balance = Balance - Amount  
92         WHERE AccountID = FromAccount;  
93  
94         UPDATE Accounts  
95         SET Balance = Balance + Amount  
96         WHERE AccountID = ToAccount;  
97
```

Query 1 Administration - Server Status SQL File 3* x

Limit to 1000 rows

```
91      SET Balance = Balance - Amount
92      WHERE AccountID = FromAccount;
93
94      UPDATE Accounts
95      SET Balance = Balance + Amount
96      WHERE AccountID = ToAccount;
97
98      SELECT 'Transfer Successful' AS Message;
99
100     ELSE
101
102      SELECT 'Insufficient Balance' AS Message;
103
```

Query 1 Administration - Server Status SQL File 3* x

Limit to 1000 rows

```
100     ELSE
101
102      SELECT 'Insufficient Balance' AS Message;
103
104     END IF;
105
106   END //
107
108   DELIMITER ;
109
110 • CALL TransferFunds(101,102,2000);
111
112 • SELECT * FROM Accounts;
```

Calling the procedure and displaying

```
110 • CALL TransferFunds(101,102,2000);  
111  
112 • SELECT * FROM Accounts;
```

Result Grid

  Filter Rows:

Edit:    Export/Import

	AccountID	CustomerID	AccountType	Balance
▶	101	1	Savings	8100.00
	102	2	Savings	17150.00
	103	3	Current	8000.00
	104	4	Savings	25250.00
	105	5	Savings	12120.00
✱	NULL	NULL	NULL	NULL